

HARI HRITHIK RA

AI/ML Engineer • Generative AI • Python • Backend Engineering

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PROFESSIONAL SUMMARY

2025 B.Tech graduate in Artificial Intelligence and Data Science with 2 peer-reviewed conference publications in multimodal AI and computer vision. Independently built and deployed four AI/ML applications (live on Streamlit and Render) and engineered deep learning architectures for quantitative financial time-series forecasting. Hands-on expertise across PyTorch, FastAPI, LLMs, RAG, computer vision, and Docker with automated CI/CD testing. Seeking a Junior AI/ML, Generative AI, or Python/Backend Engineer role.

TECHNICAL SKILLS

Languages: Python, SQL

AI / ML & Deep Learning: Machine Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Computer Vision, Time-Series Modeling, PyTorch, Hugging Face, Sentence Transformers

CV & NLP Tools: YOLO-World, spaCy, OpenCV, Stable Diffusion, LLaVA, Supervision, ReportLab

Data & Analytics: Data Science, Data Analysis, Statistical Analysis, Data Visualization, Pandas, NumPy, SciPy, Scikit-learn, Plotly

Backend & Cloud: FastAPI, REST APIs, WebSocket, OAuth2, JWT, SQLAlchemy 2.0, Alembic, PostgreSQL, SQLite, Docker, GitHub Actions (CI/CD), Pytest, Streamlit, React, Argon2id

PROJECTS

AI Hiring Assistant — Automated ATS & Talent Intelligence Engine [\[Live Demo: harihrithik-ai-hiring-assistant.streamlit.app\]](#)

Stack: Python • PyTorch • Hugging Face • Sentence-Transformers • spaCy • Streamlit • Plotly • ReportLab • Pytest • GitHub Actions

- Architected a hybrid ATS scoring engine combining 384-dimensional dense vector embeddings (all-MiniLM-L6-v2) with a curated 250+ skill taxonomy and spaCy lemmatization for contextual candidate evaluation.
- Engineered local SLM inference utilizing quantized Qwen2.5-0.5B-Instruct on CPU (torch.float16) with multi-rule deterministic validation to eliminate hallucinations and persona drift.
- Built an in-memory document compilation engine using ReportLab to generate structured, 2-page candidate assessment reports via byte streams with zero disk I/O bottlenecks.
- Implemented an automated CI/CD pipeline with 25 unit tests in Pytest and GitHub Actions, verifying deterministic scoring bounds and multi-format (PDF/DOCX) ingestion resilience.

Clarity — AI Knowledge Workspace [\[Live Demo: clarity-ai-knowledge-workspace-ctubssksgzly4wrfbcz6pn.streamlit.app\]](#)

Stack: Python • NLP • Information Retrieval • Semantic Search • Streamlit • spaCy • Scikit-learn

- Designed an offline-first, CPU-based document intelligence system ingesting PDF, DOCX, TXT, and CSV files into a searchable knowledge base with no external API dependencies.
- Engineered a five-stage NLP pipeline: text extraction → TF-IDF classification → extractive summarization → spaCy NER → 384-dimensional semantic indexing with cosine similarity and TF-IDF fallback.
- Added per-document analytics, named entity extraction, and one-click executive HTML report generation.

AI Business Intelligence Studio [\[Live Demo: ai-business-intelligence-studio.onrender.com\]](#)

Stack: Python • FastAPI • Scikit-Learn • Pandas • NumPy • SciPy • Plotly • React • Pytest • GitHub Actions

- Engineered an automated statistical BI platform that computes a composite Data Quality Index (0–100) and detects Z-score anomalies ($|z| > 3$) and Pearson correlations ($|r| > 0.70$) from tabular uploads.
- Built an interpretable AutoML Lite pipeline with StandardScaler normalization, LabelEncoding, and cross-validated model selection (Ridge, Logistic Regression, Random Forest) to isolate key drivers in <50ms.
- Architected a sub-second PDF generation pipeline streaming Matplotlib Agg byte buffers into pure-Python PDF rendering, eliminating 400MB headless browser memory overhead.
- Implemented 13 automated unit/integration tests with Pytest and GitHub Actions CI; deployed FastAPI to serve both the REST API and compiled React SPA on Render.

TaskFlow — Asynchronous Work Execution Platform & REST API [Live Demo: taskflow-api.streamlit.app]

Stack: Python • FastAPI • PostgreSQL • SQLite • SQLAlchemy 2.0 • Alembic • OAuth2 JWT • Argon2id • Docker • Pytest • Streamlit

- Architected a 3-tier REST API using FastAPI, SQLAlchemy 2.0, and PostgreSQL, decoupling routing, business logic, and database persistence using the Repository Pattern.
- Implemented OWASP-recommended Argon2id password hashing, stateless OAuth2 JWT authorization, and token-bucket rate limiting to prevent brute-force attacks.
- Engineered database lifecycle management with Alembic schema migrations and enforced multi-tenant user isolation across all CRUD operations.
- Built an automated Pytest test suite running against isolated in-memory SQLite instances in GitHub Actions CI, alongside an interactive Streamlit execution dashboard.

QUANTITATIVE ML & TECHNICAL DEVELOPMENT

Quantitative ML Developer — Algorithmic Trading Systems | Self-Directed • Ongoing

- Architected a 16-channel time-series encoding pipeline transforming OHLC market data into Gramian Angular Fields (Q-GASF/GADF), 5-bin Markov Transition Fields (MTF), and Volatility-Adaptive Recurrence Plots.
- Engineered a hybrid CNN-LSTM-Transformer Fusion Network with 4-head cross-attention fusing spatial price topologies with 30+ sequential technical indicators (SuperTrend suites and 22-period EMA ribbons).
- Implemented mixed-precision CUDA training (torch.amp), zero-copy shared memory tensor pipelines, and weighted cross-entropy loss to predict intraday directional regimes on NIFTY index data.

PUBLICATIONS

AI-Powered CCTV Analyzer: Real-Time Surveillance Analysis with Automated Reporting and Visual Analytics

International Conference on Computational Intelligence and Smart Systems (ICCISS 2025) • April 2025

- Built an open-vocabulary video analytics system using YOLO-World zero-shot detection, adaptive day/night contrast normalization, and nearest-neighbor centroid tracking.
- Implemented 2D spatial density heatmaps, rule-based threat assessment across zoned layouts, and automated PDF forensic reports with timestamped snapshots.

An AI-Powered Interactive Assistant

International Conference on Emerging Research in Computing and Systems (ICERCS 2024) • December 2024

- Designed a unified multimodal assistant integrating Stable Diffusion (v1-4), LLaVA vision-language models, and Gemma 2 with real-time speech I/O and OS-level system automation in Gradio.
- Engineered a document intelligence engine combining chunked BART-large summarization with DistilBERT SQuAD extractive question answering.

EDUCATION

Sona College of Technology • Salem, Tamil Nadu

B.Tech in Artificial Intelligence and Data Science

2021 – 2025 • CGPA: 6.71

CERTIFICATIONS

IBM SkillsBuild — Generative AI Essentials: Using LLMs to Work with Data

Issued: Aug 2026 • Credential ID: PLAN-D0F12976D81A

IBM SkillsBuild — Retrieval-Augmented Generation for Enhanced AI Outputs

Issued: Aug 2026 • Credential ID: PLAN-EA9A4259F58B

LIVEWIRE India — Python Programming

Issued: Jun 2019 • Credential ID: LWR190408008

ACHIEVEMENTS

2nd Place — Vervathon Hackathon

Department of Information Technology, Sona College of Technology • November 2023